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Cross-sensor evaluation of the model fine-tuned on Sentinel-2 against the Φ -sat-2 test split gives an mIoU of 0.11 (Base), which is consistent with the domain gap measured in Chapter 4

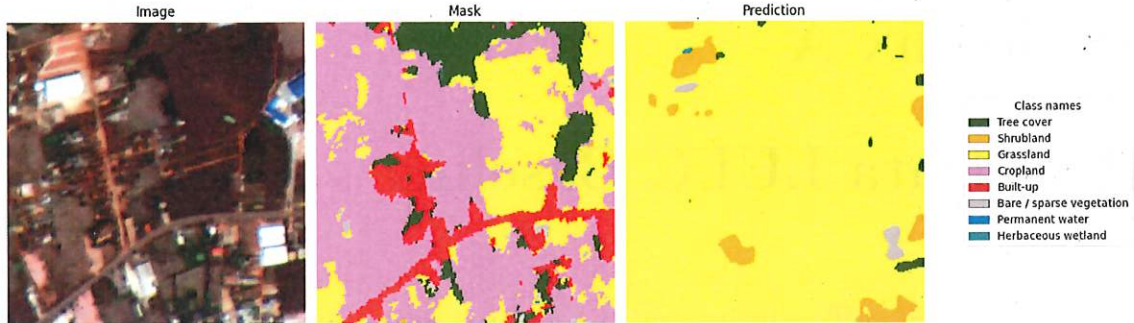


Figure A.1: Sample prediction from real Φ -sat-2 test set of the TerraMind Base model fine-tuned on Sentinel-2. The model is unable to generalize to the new sensor and predicts grassland for the entire scene, which is the most common class in the training set.

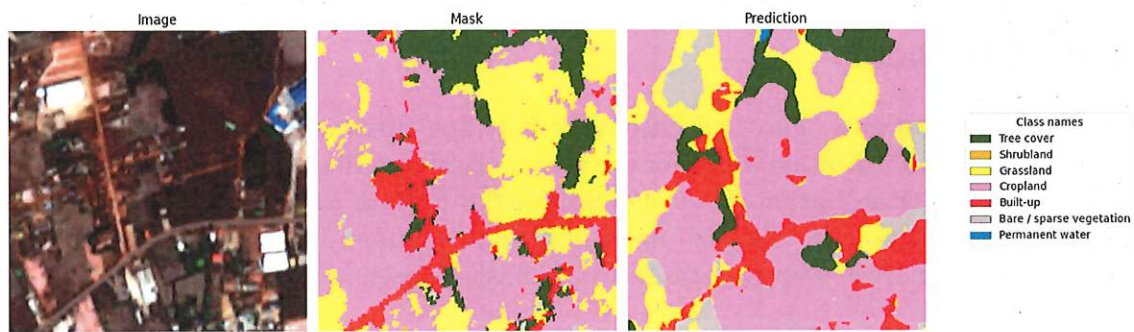


Figure A.2: Sample prediction from the Sentinel-2 patch of the same scene as of Figure A.1 from the TerraMind Base model fine-tuned on Sentinel-2. The model is able to correctly predict the classes of the scene, which shows that the poor performance on Φ -sat-2 is not due to a difficult scene or a poor model, but rather to the domain gap between the two sensors.

Declarations

Collaboration and Access to Unpublished Work

This thesis was carried out in collaboration with the European Space Agency Φ -lab, which provided access to real Φ -sat-2 L1C-L1A acquisitions via the Insula Perception platform.

Φ satNet is cited from an unpublished draft manuscript made available by its authors and consulted with their permission. That draft is not publicly available at the time of writing, and the statements this thesis stands on reflect the state of that draft at the time it was consulted. The model's codebase and weights are publicly available at on the main author's [GitHub repository](#)

All the research design, implementation, experimental work, analysis, and interpretation of results are the author's own. No text was incorporated without the author's review and revision, and the author takes full responsibility for the entire content of this thesis, including any material developed with an LLM assistance.

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Chapter 5

Conclusion

On-board processing exists to shorten the path from the acquisition to the actionable information, for time-critical Earth observation applications such as flood and wildfire response, where the value of a detection decays with every hour spent waiting for a ground-based product. Because the data of a mission’s own sensor cannot exist before its launch, and because labeled examples of rare events accumulate slowly afterward, foundation models pre-trained on the archives of other missions are an attractive way to reach a useful performance with few mission-specific labels. This interest is conditional, however, on the question this thesis set out to answer directly: whether the learned representation of such a model survives contact with a sensor it was never trained on, and, if it does not, whether this can be fixed.

The evidence we gathered here says that it can, but not for free, and not through the mechanism this thesis originally expected. TerraMind’s latent representation of real Φ -sat-2 imagery does differ measurably from its representation of Sentinel-2 imagery of the same scenes. The two sensor views of the same ground are close to orthogonal in its embedding space, and a linear probe transfers only a third of its source-domain accuracy to the target sensor. This separation is present from the very first layer of the encoder, which points to a low-level, radiometric origin, consistent with the differing spectral response between the two sensors, and not to a semantic effect, which would be expected to emerge only in the deeper layers.

Naively fine-tuning on physics-simulated degraded data does not close this gap. An explicit domain-adaptation mechanism is required, and the co-registered sensor pairs collected for this thesis are what made it possible to test the candidate mechanisms against real data, instead of assuming that one of them would work. The mechanism that did close the gap, which is the domain-specific batch normalization, only needs to know which sensor produced a given input, and not that the inputs be paired across sensors. A paired-contrastive objective exploiting the pairing directly was also implemented, and it added nothing measurable beyond it on the **LULC** task. Both mechanisms were applied during the knowledge distillation, which is the same step that is separately required to compress the model onto space-qualified hardware that cannot run its original Vision Transformer architecture at all. The deployable student is therefore domain-adaptive by construction, and not by a hoped-for inheritance from the teacher.

To answer whether this adaptation is worth implementing, we compared the distilled, domain-adapted student against a supervised baseline of identical architecture trained directly on the target sensor. Distilling with domain-specific normalization outperforms a supervised baseline at a representative label budget, but it loses to that same baseline

margin similar to the one on the target (+0.116 against +0.185 test mIoU relative to the preceding configuration). A mechanism that only aligned domains should have improved the target branch alone. The most likely explanation is that the train/test normalization inconsistency described in Section 4.1 the prior configurations were evaluated through a normalization path that the encoder had never been optimised under, which depressed the two branches at once. On that reading, part of what is reported here as a domain-adaptation gain is the repair of a latent defect in the two-branch training scheme. That caveat does not diminish the practical result, but it does bear on how much of it should be attributed to DA

4.7.2 Limitations

All experiments use at most 10,000 of the 202,582 available training patches, under 5%, so the crossover of Section 4.6 is characterised entirely within the low-data regime, and the ranking at full scale is untested. Single seeds are used everywhere except the degradation-ladder evaluation of Section 4.3, so apart from that section no result in this chapter is supported by independent replication; epoch-to-epoch validation variance is of the order of 0.02 mIoU, and differences below roughly 0.03, which includes the small-budget end of the crossover in Section 4.6, are consequently not resolved. Two further controls are outstanding at the time of writing: a shared-normalization distillation run isolating the contribution of domain-specific normalization (Section 4.1), and a recomputation of the invariance measures under shared and per-domain-standardised normalization (Section 4.4). The student is trained from a random initialisation because no pre-trained weights exist for this backbone, and this plausibly accounts for a substantial share of the remaining 0.15 mIoU gap to the teacher, and it confounds any claim that this gap is attributable to a domain shift. Finally, the work is scoped to a single sensor pair, so the finding that the gap is radiometric rather than resolution-driven (Section 4.2) should not be generalised beyond it.

4.7.3 Interpretation of the flood-robustness result

The flood-IoU degradation observed on TerraMind across the simulated degradation ladder (Section 4.3) at worst -0.152 IoU under full-severity noise and -0.008 under full-severity blur is expected. Du et al. [15] independently report that flood detection was anomalously robust to the sensor domain gap on Kanyini, in contrast to an approximately 47% drop in cloud-detection performance on the same platform. We observe the same pattern, and a plausible explanation is that the strong near-infrared absorption contrast of water survives the sensor noise and the band mismatch much better than the texture-dependent and boundary-dependent cues that cloud detection or general land-cover classification rely on. Flood segmentation may then understate the severity of the domain-gap problem.

4.7.4 Positioning with Φ satNet

Φ satNet (Section 2.6) reports that a purpose-built, non-transformer model trained from scratch on a mission-specific corpus matches or exceeds unconstrained transformer-based GeoFMs on several on-board tasks [21]. This raises a central operational question: if a

since Section 4.4 shows that the student performs similarly on the two domains, so what separates it from the teacher is its capacity and its initialisation.

Complexity against accuracy. The comparison above holds the architecture fixed and varies only the training signal, but for a space-constrained deployment the question is how much accuracy can be gained for a given parameter budget. Φ satNet has a clear advantage at only 351 k parameters compared to the U-Net that both methods use, and it is the class of model a mission would actually carry on board. Table 4.7 compares Φ satNet trained on the same 10,000 labelled patches and configuration as the supervised baseline, from two initialisations: the released weights for land-cover pre-training, and from scratch.

Model	Encoder	Decoder + head	Total	Test mIoU
Distilled student (KD + DSBN)	4.72 M	5.14 M	9.86 M	0.4481
Supervised baseline U-Net	4.72 M	5.14 M	9.85 M	0.3739
Φ satNet, pre-trained	199 k	152 k	351 k	0.3316
Φ satNet + KD + DSBN	199 k	152 k	352 k	0.3268
Φ satNet, random init	199 k	152 k	351 k	0.3215

Table 4.7: Trainable parameters against accuracy at 10,000 labelled patches. Every parameter is trained in all five rows, so the totals are also the trainable counts. Test mIoU is on the full 25,323-patch test split.

Φ satNet drops to 0.042 test mIoU compared to the supervised baseline and 0.117 against the distilled student, but it has 28.1 times fewer parameters. The distilled student's own margin is bought for a small parameter cost: domain-specific normalization adds 6,848 parameters to the shared architecture. Section 4.7.1 treats how much this mechanism contributes to the accuracy gain.

Running the distillation of Section 4.5 with the same settings (TerraMind teacher) but with Φ satNet in place of the U-Net student gives 0.3268 test mIoU, which is 0.005 below the same architecture trained supervised and 0.005 above it from a random initialisation. All three Φ satNet configurations sit inside a band of 0.01, which is statistically insignificant.

A 0.35 M student has no representational room left for the teacher's soft targets to occupy once the task loss has claimed it, so the additional supervision has nothing to buy. The per-class pattern supports this, because the U-Net's distillation gain is concentrated in the rare tail and that is precisely where the small student cannot follow: on snow and ice the distilled U-Net reaches 0.523 against Φ satNet's 0.248, on mangroves 0.341 against 0.094, and on moss and lichen 0.263 against 0.001. On the common classes the two are close, and on permanent water the small model is marginally ahead (0.634 against 0.631). This is the compression gap of Section 4.5 measured rather than asserted: the residual between teacher and student does not shrink when the teacher's signal is handed to a smaller student, because what separates them is capacity and not supervision.

Fine-tuning from the released land-cover checkpoint, pre-trained on simulated Φ -sat-2 imagery, beats the same architecture from a random initialisation by 0.0102 test mIoU. The two Φ satNet rows differ only in their initialisation, and they show that the mission-specific pre-training does not buy a measurable advantage on this task. Two further

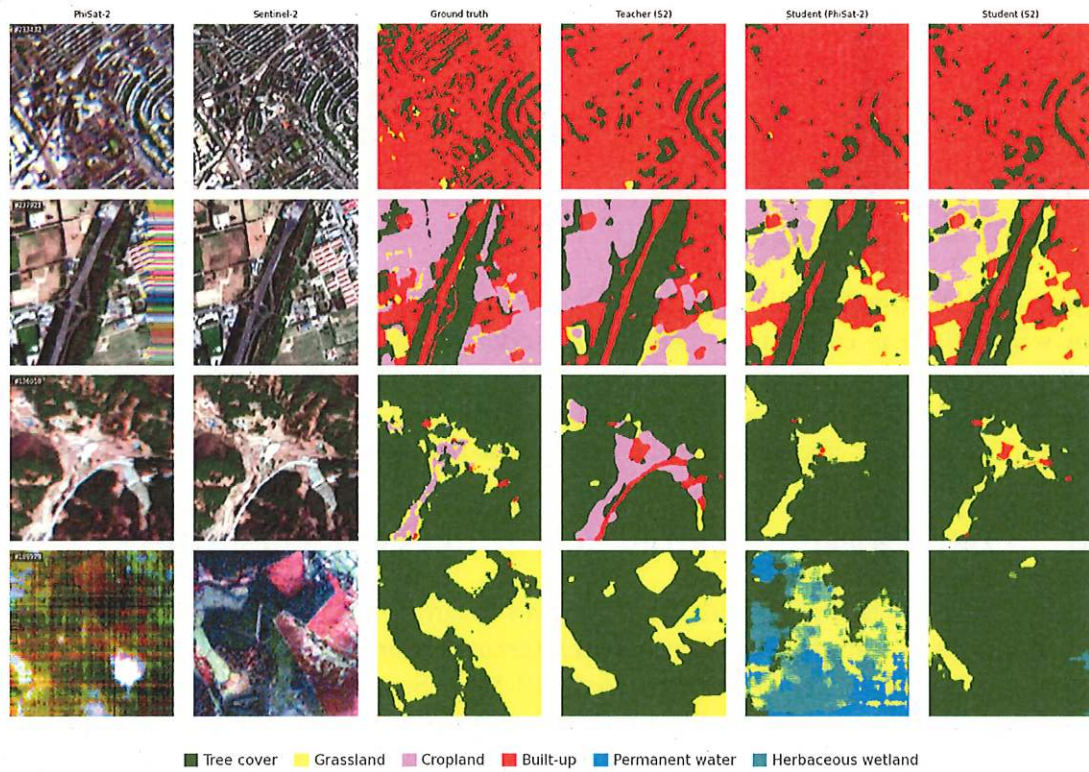


Figure 4.5: Four patches from the validation set of the KD + DSBN run, at epoch 98 of 101. Left to right: the real Φ -sat-2, its Sentinel-2 co-registered view, WorldCover label map, frozen teacher evaluated on Sentinel-2, and the student’s two branches. The teacher sees only the Sentinel-2 view, and one teacher output supervises both student branches (Figure 3.17).

Measure	TerraMind teacher	Distilled student
Paired cosine similarity	0.098	0.948
Median paired link (UMAP units)	6.15	0.40
Probe: Sentinel-2 $\rightarrow$ Sentinel-2	0.754	0.663
Probe: Sentinel-2 $\rightarrow$ Φ -sat-2	0.250	0.646
Probe transfer retention	33%	97%
IoU(Φ -sat-2) – IoU(Sentinel-2), test	—	–0.004

Table 4.5: Sensor invariance of the teacher and the distilled student, over 800 co-registered test patches. Paired cosine and probe transfer compare the two sensor views of the same ground. Probe transfer fits a linear classifier on Sentinel-2 embeddings and applies it to Φ -sat-2 embeddings; retention is the ratio of the two accuracies. The IoU difference is measured on the full test split.

against 1.955 for the teacher), which means that a linear probe separates the two domains perfectly. This is an artefact of the mechanism itself, since the domain-specific normalization gives each sensor its own affine transform. A constant per-channel offset distinguishes the domains by construction, and a linear probe finds it trivially regardless of how well the semantic content is aligned. Standardising each domain on its own statistics removes the first and second moments that a per-domain affine can express, and it drops the probe to chance for both models. Figure 4.4 shows the underlying quantity: in the raw setting the two domains’ probe scores are separated by an empty margin, and after per-domain standardisation they collapse onto one another.

PAD is therefore not a valid measurement for a model that carries a domain-specific normalization, even though it is scale-free and otherwise well suited to comparing representations of different dimensionality. The probe transfer, which asks the functional question directly, is the appropriate measure here, and it is the one we report above.

With a domain-specific normalization, the encoder is domain-adaptive rather than domain-invariant: the convolutional weights are shared but the normalization statistics are not, and the target-sensor branch has to be selected at inference. It follows that after per-domain normalization, the structure learned on the source sensor is 97% readable on the target one. That is weaker than a single sensor-agnostic encoder, but it is the property that an on-board deployment requires, since the acquiring instrument is always known.

Figure 4.5 shows what that parity looks like on individual patches. In the first three rows the two student columns are close to interchangeable: the same field boundaries, the same built-up extent, the same forest fragments, produced from two sensors whose raw digital numbers do not even share a scale (Section 3.1.1). The student also recovers finer structure than the teacher does on the first row, where the teacher’s map smooths the ridged vegetation pattern that both student branches keep. The fourth row shows a failure case with a noisy real acquisition, making the Φ -sat-2 branch collapse to permanent water and herbaceous wetland across most of the patch, while the Sentinel-2 branch of the same weights stays close to the label map.

	Distilled student (encoder frozen)	TerraMind base (encoder frozen)	TerraMind base (fine-tuned)
Clean baseline (IoU)	0.814 ± 0.002	0.739 ± 0.057	0.773 ± 0.033
<i>Sensor noise</i> (SNR)			
L1	-0.031 ± 0.024	$+0.002 \pm 0.003$	-0.001 ± 0.001
L2	-0.114 ± 0.099	$+0.003 \pm 0.006$	-0.004 ± 0.002
L3	-0.280 ± 0.234	-0.018 ± 0.017	-0.041 ± 0.014
L4	-0.458 ± 0.267	-0.088 ± 0.066	-0.152 ± 0.030
<i>Optical blur</i> (PSF)			
L1	-0.002 ± 0.000	-0.004 ± 0.003	0.000 ± 0.000
L2	-0.013 ± 0.003	-0.010 ± 0.007	0.000 ± 0.001
L3	-0.048 ± 0.014	-0.016 ± 0.012	0.000 ± 0.003
L4	-0.110 ± 0.030	-0.025 ± 0.016	-0.008 ± 0.009
<i>Band misalignment</i>			
L1	-0.006 ± 0.001	-0.008 ± 0.003	-0.003 ± 0.001
L2	-0.031 ± 0.001	-0.033 ± 0.005	-0.013 ± 0.003
L3	-0.082 ± 0.004	-0.065 ± 0.005	-0.051 ± 0.007
L4	-0.140 ± 0.007	-0.119 ± 0.009	-0.109 ± 0.012

Table 4.3: Domain gap across the degradation ladder: change in positive-class IoU relative to the clean test split, for a model trained only on clean data. Values are the mean over three seeds of the within-seed drop, $\pm$ one standard deviation of those drops.

the shift has already destroyed.

Sensor noise produces the largest single degradation, but its magnitude is a property of the model rather than of the perturbation, and it is the least reproducible quantity in the table. At full severity it costs the distilled student -0.458 IoU on average and fine-tuned TerraMind -0.152 , a threefold difference that holds at every severity.

Optical blur is the least damaging factor in every configuration, but the three differ in how cleanly the effect is resolved. For fine-tuned TerraMind it is indistinguishable from zero: four severities spanning a twenty-four-fold reduction in image gradient energy move the score by at most 0.008 IoU. For frozen TerraMind the response is monotone and uniformly negative but never clears the significance threshold, reaching -0.025 at full severity. For the distilled student it is small but unambiguous, significant at all four severities with the tightest deviations in the table, rising from -0.002 to -0.110 . Blur is therefore real for the student and negligible for TerraMind, yet even for the student it remains the smallest of the three factors, costing a quarter of what noise costs at the same nominal severity.

This ordering of factors resolves the attribution ambiguity left open in Section 4.2. The factor that models a resolution difference, which is the optical blur, is the one that costs almost nothing, while the radiometric factor and the band-registration factor are the ones carrying the effect. The per-layer PAD result located the gap in low-level radiometry rather than in semantics, and the ladder independently identifies the two low-level physical effects responsible, using a different task, a different measurement, and imagery in which scene content is held fixed by construction. It also reinforces the point that the gap here is not resolution-driven, which is again the reverse of the emphasis

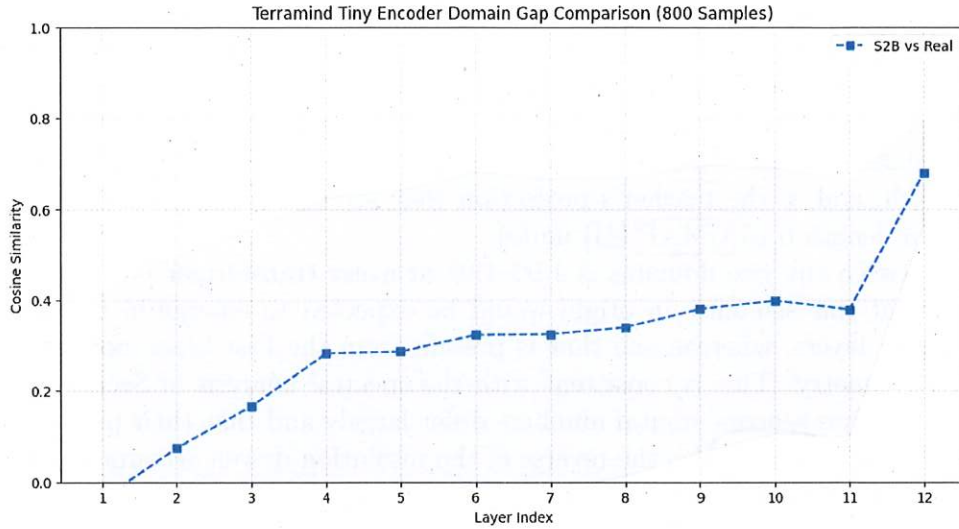


Figure 4.2: Per-layer cosine similarity between raw paired Sentinel-2B and real- Φ -sat-2 patch embeddings from the TerraMind Tiny encoder, averaged over 800 samples (shaded band: one standard deviation).

4.3 Degradation effects on Downstream Task Gap

This section takes up the last part of the previous research question, which asks which of the physical differences between the two instruments is the one that produces the gap. It builds on the measurements above, which established that a gap exists and that it is low-level in origin, and narrows the question to three candidates: radiometric noise, optical blur, and band misalignment. The two instruments differ in their spectral response, their radiometric noise, their band registration and their ground sampling distance all at once, and no measurement on real pairs can separate those. The simulated ladder of Section 3.1.2 makes the separation possible by degrading one factor at a time over identical scene content. Three factors are varied over four severities each: sensor noise (SNR), optical blur (PSF), and band-to-band misalignment. Table 4.2 gives the parameter values. A severity axis common to all three is defined by setting level 4 to the operating point of the joint simulator configuration and halving the perturbation amount at each step down, so that a given level means the same fraction of full simulated degradation whichever factor is being varied.

Level	Severity	SNR range	PSF σ (px)	Misalignment σ (px)
L1	0.125	[40, 80]	0.5	1.25
L2	0.250	[20, 40]	1.0	2.50
L3	0.500	[10, 20]	2.0	5.00
L4	1.000	[5, 10]	4.0	10.00

Table 4.2: The degradation ladder. Level 4 reproduces the joint simulator configuration; each step down halves the perturbation. SNR is drawn per scene from the stated range and the noise amplitude scales as its reciprocal, so the four ranges correspond to relative noise of 1/8, 1/4, 1/2 and 1. Blur and misalignment parameters are in resampled 4.75 m pixels.

The contrastive representation-alignment objectives (Section 3.4.3) were implemented and ablated, but they are disabled in all the results below, since their weights are zero throughout. That ablation is reported separately.

Table 4.1 lists the settings of both methods. Everything the two conditions could share is held constant: architecture, initialisation, split, seed, normalization, augmentation, task loss, optimiser, schedule, and early-stopping criterion are identical, and the methods differ only in the presence of the teacher, the second sensor branch, and domain-specific normalization.

Setting	KD + DSBN	Supervised baseline
<i>Shared</i>		
Student architecture	U-Net, 7 input bands, 32 decoder channels, 1×1 conv head	
Initialisation	Random (no pre-trained weights available)	
Optimiser	AdamW	
Learning rate	1×10^{-4}	
Weight decay	1×10^{-4}	
LR schedule	ReduceLROnPlateau, factor 0.5, patience 5	
Batch size	8	
Early stopping	Patience 12 on target-sensor validation IoU	
Model selection	Best target-sensor validation IoU	
Task loss	Focal, $\gamma = 2.0$	
Class weighting	Inverse square-root of measured pixel frequency	
Augmentation	Joint geometric (flips, 90° rotations)	
Random seed	42	
Numerical precision	32-bit	
<i>Distillation-specific</i>		
Teacher	TerraMind, frozen	—
Distillation loss	KL on logits, $T = 4.0$	—
Distillation applied to	All pixels, both branches	—
Task-loss weight	1.0 target + 1.0 source	1.0 target
Distillation weight	1.0	—
Contrastive weights	0 (instance and pixel)	—
Domain-specific BN	Enabled	—
Sensor branches	Two (Φ -sat-2, Sentinel-2)	One (Φ -sat-2)

Table 4.1: Training configuration for the two methods. Settings above the rule are identical by construction; those below apply to the distilled method only.

4.2 Sensor Gap in TerraMind Latent Space

The first question is whether the domain gap is visible in the representation itself, and not only in the downstream accuracy. We extracted embeddings for 800 co-registered test patches from both sensors, and compared them under three measures: the paired cosine similarity between the two views of the same ground, the proxy $\mathcal{A}$ -distance (PAD) [13], and a cross-domain probe-transfer test in which a linear classifier fitted on Sentinel-2 embeddings is applied to the Φ -sat-2 embeddings of the same patches.

3.6 Evaluation Protocol

Four evaluation methods follow from the mechanisms described, and they correspond to the three research questions of Section 1.3. First, whether the sensor gap is visible in the teacher’s representation is measured by paired cosine similarity, per-layer PAD, and cross-domain linear-probe transfer between the two input domains. Second, whether the distilled student’s representation inverts these measurements relative to the teacher, and which of the two candidate mechanisms is effective (DSBN, contrastive alignment). Third, whether distillation with the adopted mechanism improves LULC task performance over a supervised baseline of identical student architecture trained directly on labelled Φ -sat-2 with no teacher and no second sensor, at a label budget large enough to be representative and, separately, at a much smaller budget, since the direction of this comparison is not assumed to be constant across label budgets. Fourth, flood segmentation across the simulated degradation ladder, to characterize how gracefully performance degrades with increasing sensor-noise severity. A few-shot label-efficiency comparison against Φ satNet (Section 2.6) on matched small label budgets, testing directly whether adapting TerraMind’s broad, diverse pre-training generalizes to Φ -sat-2 with fewer labels than Φ satNet’s mission-specific pre-training requires, remains planned but has not been run (Section 4.7.3). The earlier protocol of fully fine-tuning TerraMind directly on real Φ -sat-2 LULC labels, evaluated on the same real domain and without any cross-sensor test, produced a baseline mIoU that we retain in Appendix A for reference. It is not the main baseline of interest, however, because it does not test the cross-sensor generalization, and it therefore does not address the research questions of this thesis.

3.7 Tools, Software Environment, and Reproducibility

reference The degradation pipeline and dataset construction are implemented as modular EOTask objects within the eo-learn framework, orchestrated through a versioned simulation configuration object instead of hard-coded parameters, so that a given degradation run (baseline-calibrated or extended-sweep) is fully specified by its configuration. Triplet and LULC datasets are stored in HDF5 with an accompanying manifest recording per-patch provenance and train/validation/test split assignment (fixed random seed, positional split), enabling exact reproduction of any reported split. Model training (TerraMind fine-tuning, DSBN-based distillation, the contrastive-alignment ablation) is implemented in PyTorch. The codebase is organized as a single versioned Python package, with sub-modules separating the data simulation, the dataset loading, the model training tasks and the drift analysis, under a standard git version control, so that each experiment reported in Chapter 4 corresponds to a retrievable configuration.

available code, data, model/weights, etc?

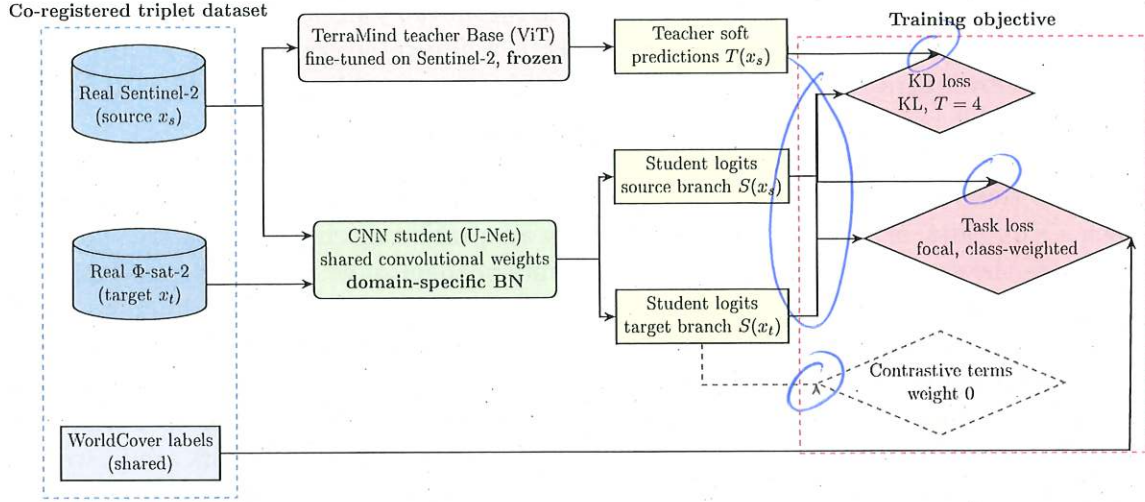


Figure 3.17: Cross-architecture distillation with domain-specific normalization. The frozen TerraMind teacher is evaluated only on the Sentinel-2 view, where it is trustworthy; its soft predictions supervise both student branches, which is possible because the two sensor views depict the same scene. The student is forwarded twice per patch, once per sensor, selecting its own batch-normalization statistics each time while sharing all convolutional weights. Both branches also receive the task loss against the single shared WorldCover label map. The contrastive terms were implemented and evaluated but carry zero weight in every result reported in Chapter 4

Distillation term. $\mathcal{L}_{\text{KD}}$ is the temperature-scaled **KL** divergence of Hinton et al. [14], applied independently at every pixel of the map $\mathcal{P}$,

$$\mathcal{L}_{\text{KD}}(z^{\text{stu}}, z^{\text{tea}}) = \frac{\tau^2}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \sum_{c=1}^C q_{ij,c} \log \frac{q_{ij,c}}{p_{ij,c}}, \quad \begin{aligned} p_{ij} &= \text{softmax}(z_{ij}^{\text{stu}} / \tau), \\ q_{ij} &= \text{softmax}(z_{ij}^{\text{tea}} / \tau), \end{aligned} \quad (3.17)$$

at temperature $\tau = 4$, with the τ^2 factor restoring gradient magnitudes to the scale they would have at $\tau = 1$. Two properties of how this term enters Equation (3.15) matter more than its functional form. First, both occurrences take the *same* second argument $T(x_s)$. The teacher is evaluated once per patch, on the Sentinel-2 view alone, and its output is detached and used to supervise both student branches, so the student’s Φ -sat-2 predictions are pulled toward a teacher output computed from the Sentinel-2 view of the same ground. This is what keeps the teacher’s own domain-shift error out of the objective, and it is available only because the two views are co-registered. Second, the sum runs over every pixel of the map, including those the WorldCover rasterization left unlabelled, which is where distillation contributes supervision that Equation (3.16) cannot.

Contrastive terms. $\mathcal{L}_{\text{inst}}$ is a symmetric NT-Xent between the pooled, ℓ_2 -normalised bottleneck embeddings of the two sensor views of a patch, at temperature 0.1, taking the co-registered view as the positive and all remaining in-batch embeddings as negatives. $\mathcal{L}_{\text{pix}}$ is its class-conditional dense counterpart, pulling each pixel embedding toward an exponential-moving-average prototype of its own class that is shared between the two domains and pushing it from the other class prototypes, similar to the dense multi-sensor alignment loss of X-STARS [29]. Both weights are zero in every result reported

Since the student must eventually run on the target hardware (Myriad 2), we also have to establish the cost of the `DSBN` mechanism and its implications for the model architecture. Batch normalization is representable in the deployment format: `BatchNormInference` belongs to `opset1`, the earliest OpenVINO operator set, and is therefore within the operator set available to the 2020.3 release that supports this processor.¹ In practice, however, the operator is not emitted at all, because at inference a batch-normalization layer reduces to a fixed per-channel affine transform. OpenVINO’s model conversion folds the transform into the weights of the preceding convolution by default. Since we are only interested in the target sensor, only the target-sensor branch is exported, and its statistics fold into the convolution weights just like a normal single-domain model. Therefore, the deployed model carries no batch-normalization operation, and `DSBN` adds neither an operator-support risk nor an inference cost relative to the supervised baseline. This argument follows from the documented conversion behaviour of the toolchain, not from an export of the student to the Φ -sat-2 processor, which has not been carried out.

3.4.3 Paired-contrastive triplet alignment

The co-registered triplet dataset provides something that most domain-adaptation settings lack, which is, for every training example, the identical underlying scene observed by two or three different sensors. This allows a directly supervised alignment objective in place of an unpaired one. The contrastive loss pulls the embeddings of matched real- Φ -sat-2, real-Sentinel-2, and simulated- Φ -sat-2 patches of the same scene together, while pushing apart embeddings of unrelated scenes, regardless of which sensor produced each embedding. This directly penalizes the encoder for representing the sensor identity and not only the scene content, and it does so with the strongest supervisory signal available to us, which is an exact scene correspondence, instead of matching marginal distributions between unpaired sensor domains as an `MMD`-style objective would.

This objective was implemented and evaluated on top of the `DSBN` configuration above, with the co-registered triplets used to test a training mechanism. On the `LULC` task, it did not improve downstream performance beyond the `DSBN`-only configuration (Section 4.4), and all headline results in Chapter 4 are reported with its weight set to zero. The co-registered pairing that makes a contrastive objective possible is an advantage of this dataset, but the fact that exploiting it directly didn’t add measurable improvement on top of simpler per-domain normalization scheme is informative about where the sensor gap lives (Section 4.2).

3.5 Cross-Architecture Knowledge Distillation

TerraMind’s `ViT`-based encoder cannot run on the Φ -sat-2 on-board processor: the Myriad 2 `VPU`’s supported operator set has no dynamic-axes support, which `ViT` inference requires regardless of model size. The compression must therefore cross architecture families, from a `ViT` teacher to a `CNN` student, instead of compressing within the `ViT` family as some prior on-board `GeoFM` work does [15]. The teacher is TerraMind in its base variant, fine-tuned on the Sentinel-2 branch of the triplet dataset for the `LULC` task and

¹https://docs.openvino.ai/2023.3/openvino_docs_ops_normalization_BatchNormInference_1.html Operator specifications are versioned by opset: the 2020.3 documentation set has been withdrawn by Intel, so the current page is cited for an operator whose opset membership predates it.

3.4.2 Domain-specific batch normalization

The mechanism we adopted only requires that the domain of each training example is known. Domain-Specific Batch Normalization (DSBN) [4] gives the student a separate batch-normalization affine transform and running statistics per sensor domain, on top of otherwise fully shared convolutional weights. Concretely, the two-branch distillation scheme (Section 3.5) forwards a Sentinel-2 batch and a Φ -sat-2 batch through the student separately at each training step, and DSBN lets each pass use the normalization statistics of its own domain, instead of a single set of statistics blended across the two.

Batch normalization. Write $x \in \mathbb{R}^{H \times W \times N}$ for the activations of a single channel over a mini-batch of N examples, indexed by spatial position (i, j) and example n . A batch-normalization layer [30] whitens those activations and rescales them with learned per-channel affine parameters γ and β :

$$\text{BN}(x[i, j, n]; \gamma, \beta) = \gamma \cdot \hat{x}[i, j, n] + \beta, \quad (3.7)$$

where

$$\hat{x}[i, j, n] = \frac{x[i, j, n] - \mu}{\sqrt{\sigma^2 + \epsilon}}, \quad (3.8)$$

and the mini-batch mean and variance are

$$\mu = \frac{\sum_n \sum_{i,j} x[i, j, n]}{N \cdot H \cdot W}, \quad \sigma^2 = \frac{\sum_n \sum_{i,j} (x[i, j, n] - \mu)^2}{N \cdot H \cdot W}, \quad (3.9)$$

with ϵ a small constant guarding against division by zero. Mini-batch statistics are unavailable at inference, so during training the layer additionally accumulates estimates of the mean and variance of the whole activation distribution, $\bar{\mu}$ and $\bar{\sigma}$, by exponential moving average with update factor α . At the t -th mini-batch,

$$\bar{\mu}^{t+1} = (1 - \alpha) \bar{\mu}^t + \alpha \mu^t, \quad (\bar{\sigma}^{t+1})^2 = (1 - \alpha) (\bar{\sigma}^t)^2 + \alpha (\sigma^t)^2, \quad (3.10)$$

these accumulated values whiten activations at test time.

Equation (3.9) is where a single-branch network fails under domain gap: the sums run over every example in the mini-batch regardless of which sensor produced it, so the running estimates of Equation (3.10) converge to a blend of the two sensors' statistics. Chang et al. make the same observation in general terms, that sharing the mean and variance across source and target is inappropriate when the domain shift is significant; the raw-scale and distribution-shape differences measured in Section 3.1.1 fit this description.

Domain-specific batch normalization. DSBN keeps one complete set of normalization parameters and statistics per domain label $d \in \{S, T\}$, here S for Sentinel-2 and T for Φ -sat-2. Writing $x_d \in \mathbb{R}^{H \times W \times N}$ for the activations of a channel belonging to domain d , the layer becomes

$$\text{DSBN}_d(x_d[i, j, n]; \gamma_d, \beta_d) = \gamma_d \cdot \hat{x}_d[i, j, n] + \beta_d, \quad (3.11)$$

with

$$\hat{x}_d[i, j, n] = \frac{x_d[i, j, n] - \mu_d}{\sqrt{\sigma_d^2 + \epsilon}}, \quad (3.12)$$

band follows

$$\|\Delta \mathbf{r}(b_n)\| \sim \mathcal{N}(\mu_{b_n}, \sigma^2), \quad \mu_{b_n} \in [0.44, 1.1] \text{ pixels}, \quad (3.3)$$

with direction sampled uniformly over $[0, 2\pi)$ and displacement accumulating across preceding bands. For the geometrically corrected L1C product used throughout this thesis, an on-board co-registration step removes this accumulation, and the residual is instead modeled as a single random displacement of each band relative to a reference band ($\mu_{b_n} = 0$), with standard deviation set higher over water ($\sigma_{\text{sea}} = 6$ pixels) than over land ($\sigma_{\text{land}} = 1$ pixel), reflecting the harder auto-registration problem posed by feature-poor sea surfaces.

Signal-to-noise degradation. The baseline formulation draws a per-band noise sample from $\mathcal{N}(0, 1)$ scaled by the ratio of a reference radiance to a specified **SNR**,

$$L_{\text{noise}}(i, j, b_n) = \frac{L_{\text{ref}}(b_n)}{\text{SNR}(b_n)} \mathcal{N}(0, 1), \quad (3.4)$$

This formulation implicitly treats noise variance as signal-independent, and matches the specified **SNR** only near L_{ref} . A signal-dependent variant, consistent with a shot-noise regime in which **SNR** scales with the square root of collected signal, rescales the reference **SNR** to an effective, radiance-dependent value and samples

$$L_{\text{noise}}(i, j, b_n) = \frac{\sqrt{L_{\text{in}}(i, j, b_n) L_{\text{ref}}(b_n)}}{\text{SNR}(b_n)} \mathcal{N}(0, 1). \quad (3.5)$$

The implementation used for the calibrated baseline degradation level applies Equation (3.5). The extended sweep pushes beyond the nominal Φ -sat-2 noise conditions, since the domain-invariance has to be stress-tested under a more severe degradation than the flight-validated parameters cover. For that sweep, a simplified variant samples a base **SNR** uniformly from a specified range and scales noise by the ratio of that **SNR** to the batch-wise maximum radiance, trading exact photometric fidelity for the ability to sweep degradation severity beyond the calibrated regime. The baseline level is flight-parameter-calibrated, and the extended sweep trades fidelity for range.

Optical blur via MTF/PSF. Optical blur is modeled through the system modulation transfer function (**MTF**), aggregating static (optics, detector), dynamic (jitter, platform instability), and smearing (motion) contributions. Assuming shift-invariant blur, imaging reduces to convolution with the system point spread function (**PSF**). Given the **MTF** amplitude at Nyquist frequency $M(b_n)$, the **MTF** is modeled as Gaussian in normalized spatial frequency k ,

$$\text{MTF}(b_n, k) = e^{\ln M(b_n) \cdot k^2}, \quad (3.6)$$

with the corresponding **PSF** obtained as its inverse Fourier transform and convolved with the radiance image. For Φ -sat-2 the average Nyquist **MTF** is $M_{\Phi}(b_n) \approx 0.05$; for Sentinel-2, $M_{\text{S-2}}(b_n) \approx 0.30$. The reference simulation applies Equation (3.6) directly with $M_{\Phi}(b_n)$, leaving the intrinsic Sentinel-2 blur uncorrected; a combined-filter form exists for the case where source blur should also be compensated, but is not required here since the intrinsic Sentinel-2 blur is small relative to Φ -sat-2's.

This pipeline is shared infrastructure within the research group: it also underlies the pre-training and benchmark datasets used by Φ satNet (Section 2.6). This thesis uses the

3.1.2 Simulated degraded flood-segmentation dataset

No real Φ -sat-2 flood dataset exists, because the mission has not yet captured and labeled a flood event; this is a common limitation of any new mission, as reported by Barretta et al. [5]. To obtain labeled data for the flood-segmentation task that motivates this thesis, we pass the Sen1Floods benchmark through the same degradation pipeline used to build the triplets, which produces a graded ladder of Φ -sat-2-like flood scenes at increasing degradation severity. This simulated dataset plays a different role from the triplets: it is the training and evaluation substrate for the flood task itself, not the substrate for measuring the domain gap, since it lacks a real- Φ -sat-2 counterpart against which to validate the simulation directly.

The ladder is produced with an updated implementation of the OrbitalAI simulator, restructured as a configurable sequence of stages inside TerraTorch so that it applies to any Sentinel-2-based dataset in that framework instead of a single acquisition. The segmentation mask is carried through the transformation with the imagery (Section 3.2, Figure 3.15). This reimplementaion makes it possible to control each physical effect individually, and this is what gives the ladder a second purpose, beyond supplying a simulated dataset with flood labels. The triplets establish that a domain gap exists in TerraMind’s representation (Section 3.3) but not which physical difference between the two instruments produces it, and the two sensors differ in several respects at once, so no measurement on real pairs can separate the causes. Degrading a fixed set of scenes one factor at a time, at graded severities within each factor, is what makes that separation possible. The scene content is identical across the ladder by construction, so any change in the response of the base TerraMind encoder is attributable to the effect being varied. Chapter 4 reports how the teacher responds across this ladder by isolating which individual factor carries the effect. This experiment complements the answer to the first research question.

3.2 Physics-Based Sensor Degradation Pipeline

The degradation pipeline adapts the simulator developed for the ESA Φ -lab OrbitalAI challenge [2]. This simulator projects labeled Sentinel-2 L1C tiles into the Φ -sat-2 sensor domain by modeling the physical differences between the two instruments as a sequence of operations in radiance space. Because these effects are defined physically on radiance, the pipeline converts source reflectance to top-of-atmosphere radiance, applies each degradation in turn, and converts back to reflectance to output a simulated L1C product. Pixel coordinates are written (i, j) and spectral band index b_n .

The implementation used here is an updated version of that simulator, summarised in Figure 3.15. Each stage draws its parameters from its own intermediate input instead of from a single fixed configuration, so an individual effect can be varied or disabled without disturbing the others, and this is what permits the factor-isolation use described in Section 3.1.2. Solar geometry and irradiance are retrieved per scene from the Copernicus Data Space OData API using the source product’s location and acquisition time, so the radiance conversion of Equation (3.2) uses each tile’s true illumination conditions.

the two sensors. The three classes shown with the most extreme bars (moss and lichen, snow and ice, mangroves) each account for under 1% of the labelled pixels in this sample, and they carry correspondingly enormous error bars, so they are reported for completeness but should not be read as robust estimates.

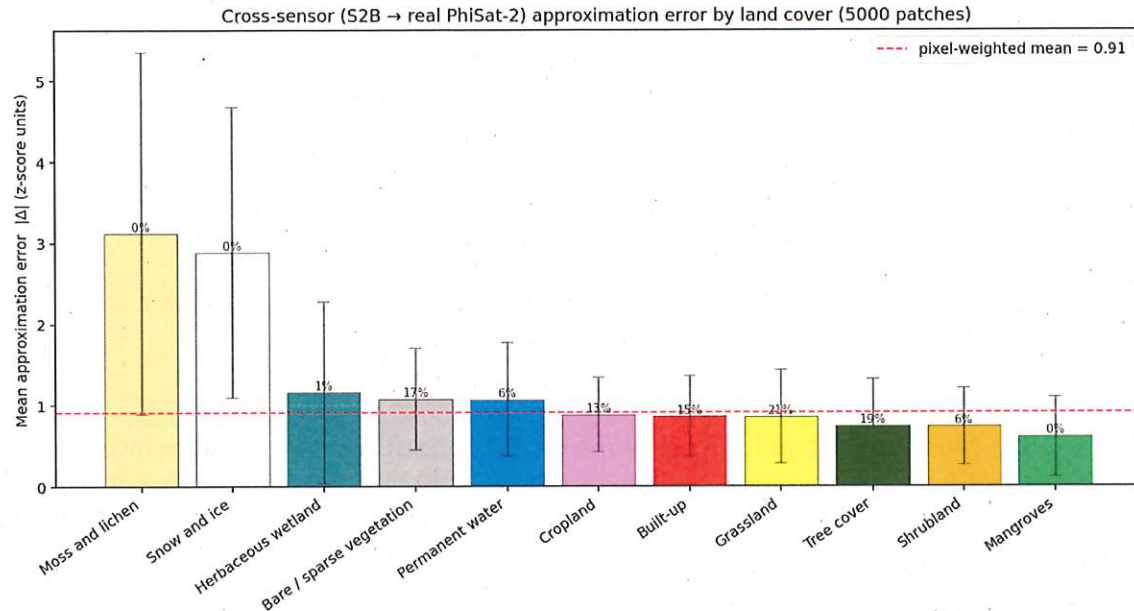


Figure 3.11: Mean cross-sensor approximation error (Equation (3.1)) by dominant ESA WorldCover class, over 5,000 patches. Error bars are one standard deviation; the percentage above each bar is that class's share of labelled pixels in the sample. Three classes show $>1\%$ presence in the sample (moss and lichen, snow and ice, mangroves) because they are rare in the dataset, they are still reported for completeness.

Qualitative examples

Figures 3.12–3.14 show three individual patches, with the LULC mask overlaid on both sensor views alongside the approximation-error map of Equation (3.1).

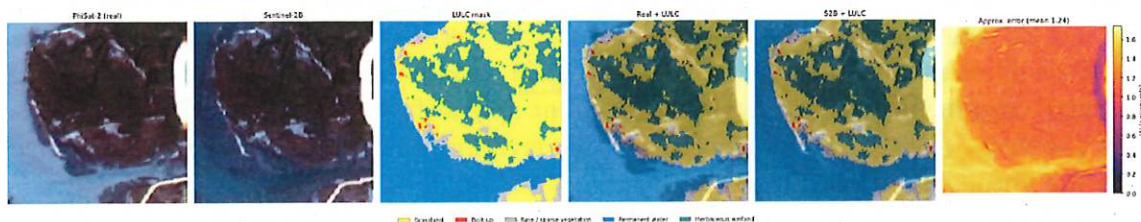


Figure 3.12: Patch showing coastal wetland (mean approximation error 1.24, above the dataset-wide mean of 0.91). Errors are located over open water, consistent with the class-level pattern of Figure 3.11.

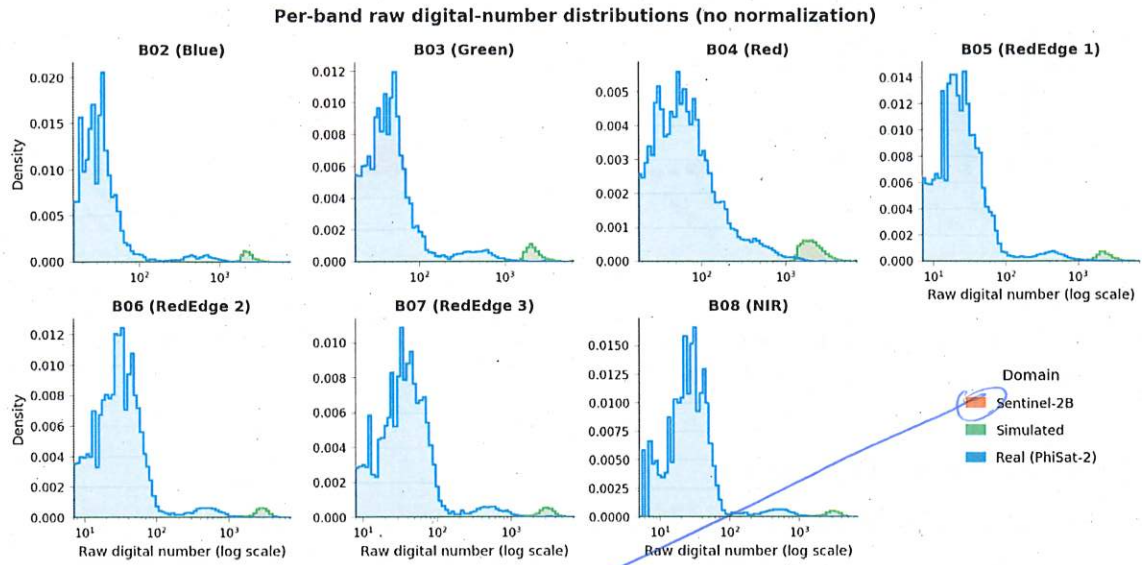


Figure 3.8: Per-band raw digital-number density by sensor (log-scaled x -axis; 400,000 sampled pixels). Sentinel-2B (orange) and simulated Φ -sat-2 (green) overlap. Real Φ -sat-2 (blue) is concentrated in lower values, with a smaller secondary lobe extending into the Sentinel-2B/simulated value range in every band.

not visible

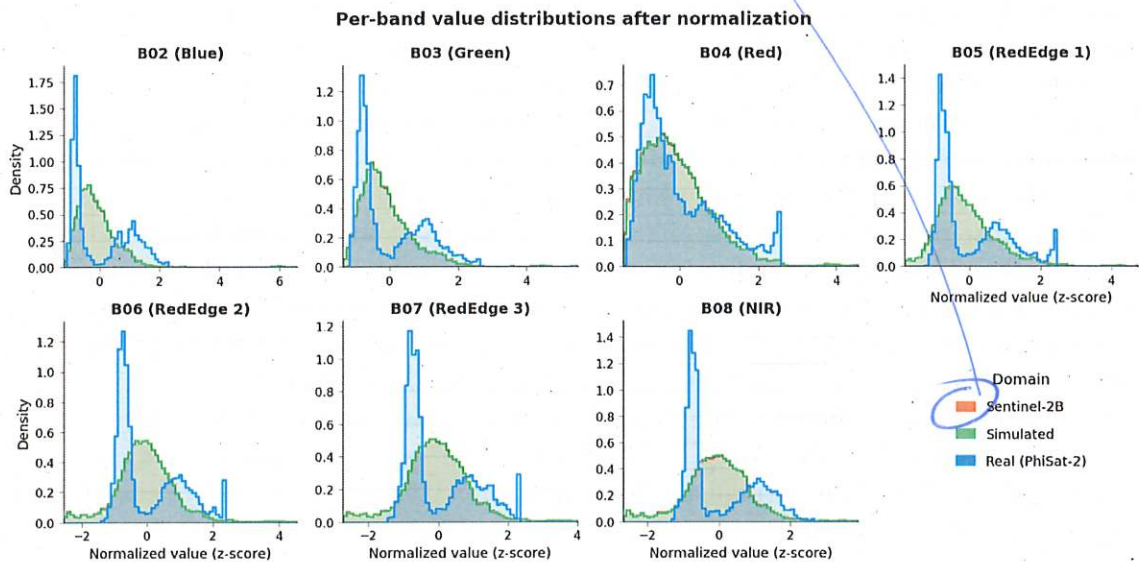


Figure 3.9: The same per-band densities after per-sensor z -score normalization of $\sqrt{DN}$.

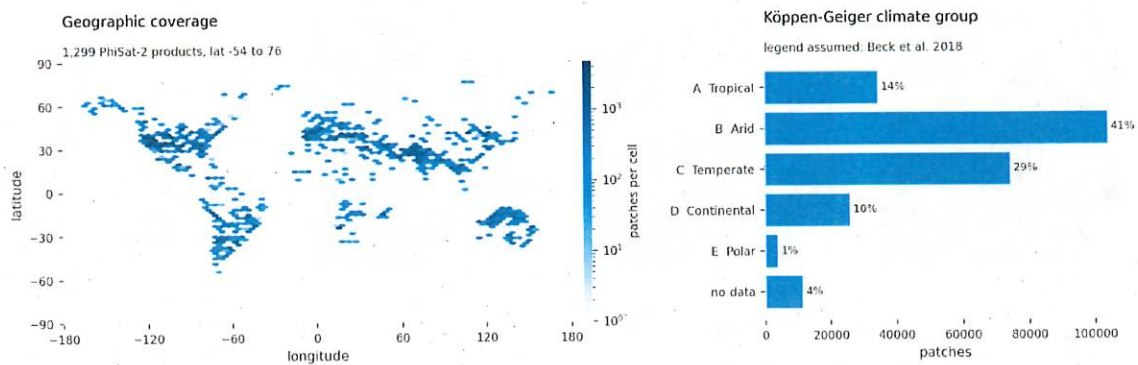


Figure 3.5: Left: spatial density of Φ -sat-2 products (hexbin, log-scaled patch count per cell). Right: distribution of patches by Köppen-Geiger climate group [3].

The eleven **ESA** WorldCover classes present are similarly imbalanced by construction: grassland (20.8%), tree cover (20.0%), bare/sparse vegetation (16.2%), built-up (14.4%), and cropland (13.4%) represent the majority of labelled pixels, while snow/ice, mangroves, and moss/lichen classes are only present at 0.6% (Figure 3.6).

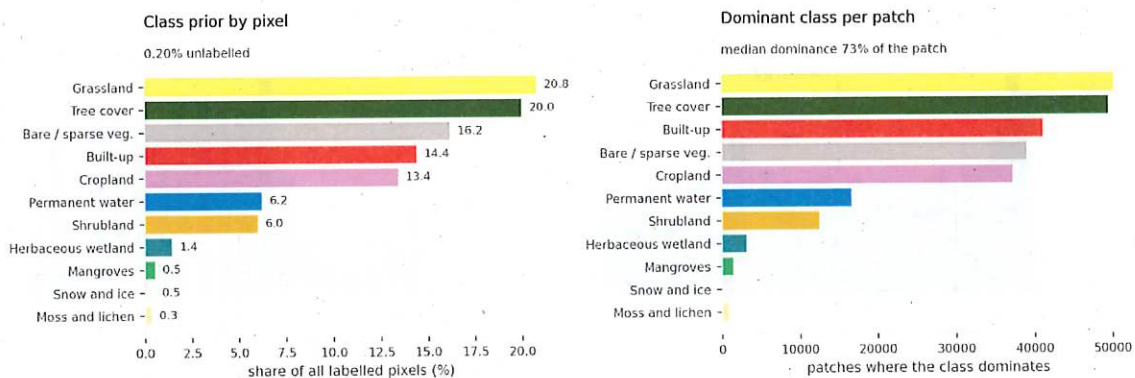


Figure 3.6: Left: share of all labelled pixels by ESA WorldCover class. Right: number of patches in which each class is the dominant (plurality) class.

Train/validation/test split

The dataset is split 80/10/10 (202,582 / 25,323 / 25,323 patches) at the patch level with a fixed seed, and the resulting class-prior distribution is stable across all three splits, confirming the split does not introduce label-distribution shift (Figure 3.7 left). However, this introduces a different problem. Because the split is made over patches, all 1,299 source products are represented in the training set, and every one of the 1,287 distinct products contributing to the test set also contributes patches to the training set (Figure 3.7 right). Since patches coming from the same product are spatially adjacent or overlapping crops of a single acquisition, the test set is therefore not a clean geographic or scene-level hold-out, and the real-data performance figures reported in Chapter 4 should be read as an optimistic upper bound on the generalization to unseen scenes. Re-splitting at product level instead of at patch level is the direct remedy, and it is noted as future work in Section 4.7

important limitation to be recalled.

2 counterpart at a comparable centre wavelength, and it is that correspondence which makes Sentinel-2 usable both as the paired view in the triplets and as the source the degradation pipeline (Section 3.2) projects into the Φ -sat-2 domain. The correspondence is close but it is not exact, and Φ -sat-2 additionally carries a panchromatic band that has no Sentinel-2 equivalent. Spatially, Φ -sat-2 samples every band at a uniform 4.75 m, whereas Sentinel-2 provides 10 m for B2, B3, B4 and B8 and 20 m for B5, B6 and B7. This spatial resolution mismatch is a factor taken into account when interpreting the domain gap in Chapter 4. The target sensor is the finer of the two, so a resolution mismatch cannot account for a loss of performance on Φ -sat-2, whereas the spectral differences visible in panel (a) remains a potential explanation.

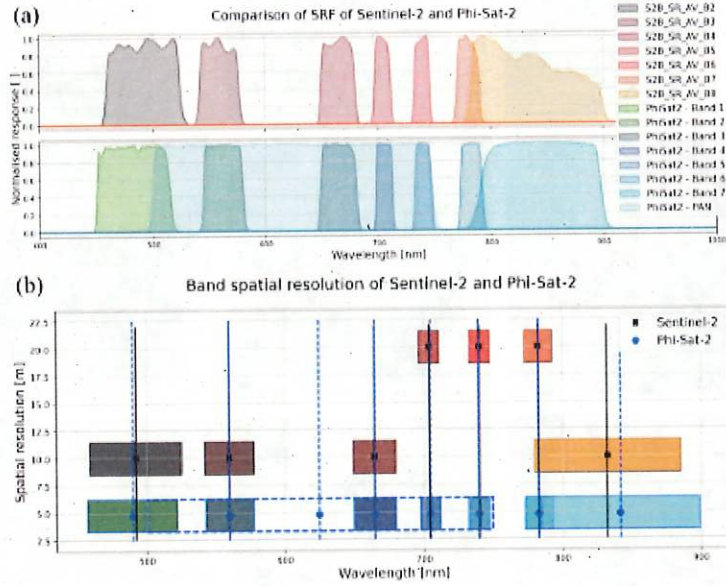


Figure 3.2: Band correspondence between Sentinel-2B and Φ -sat-2, from Longép  et al. [2]. (a) Normalized spectral response functions, Sentinel-2B above and Φ -sat-2 below. (b) Per-band spatial resolution. The correspondence in (a) makes Sentinel-2 usable as the paired source; the residual differences in both panels are the sensor gap this thesis aims to measure.

This dataset carries labels for a single task: land-use/land-cover (LULC) classification as shown in Figure 3.3. These labels are derived from ESA WorldCover 2021 [37] and were rasterized onto each triplet’s real- Φ -sat-2 footprint by the team at the Φ -lab. It is the only real-sensor ground truth available to us for any downstream task, and the co-registration allows us to measure the sensor-induced covariate shift in TerraMind’s latent space (Section 3.3). Prior domain-gap studies have been limited by the scene-content and geography confounds, comparing unrelated source and target datasets.

Chapter 3

Methodology

This chapter presents the complete analytical workflow. We first describe the data used and the role each dataset plays, with the physics-based degradation pipeline and the co-registration procedure that produces the triplets dataset. Then, the protocol for measuring the domain gap is described, with the domain-adaptation mechanism adopted and the cross-architecture distillation procedure, and finally the evaluation protocol. The chapter closes with the tooling and the reproducibility practices used throughout the process.

This work uses two distinct datasets compiled specifically for the Φ -sat-2 mission, each with a different purpose. The first is the co-registered triplet dataset. Real Φ -sat-2 acquisitions are matched to concurrent Sentinel-2 scenes using the LightGlue deep feature-matching algorithm [36]. Simulated Φ -sat-2 renderings of the Sentinel-2 scenes are also produced with the physics-based degradation pipeline described in Section 3.2. This dataset carries land-cover labels derived from ESA WorldCover 2021 [37], and its purpose is to have a downstream task with which to measure the domain gap caused by the sensor differences between Sentinel-2 and Φ -sat-2. The second dataset is a physics-degraded version of the Sen1Floods flood-segmentation benchmark, produced with the same degradation pipeline. It stands in place of a real labeled Φ -sat-2 flood dataset, which does not yet exist because the mission has not captured an annotated flood event.

3.1 Data

3.1.1 The co-registered triplet dataset

The main dataset is a collection of spatially and temporally co-registered image triplets, produced in collaboration with the ESA Φ -lab: real Φ -sat-2 acquisitions (obtained via the ESA Insula Perception platform), the corresponding real Sentinel-2B scene, and a physics-simulated Φ -sat-2 rendering of that same Sentinel-2 scene (Section 3.2). Candidate scenes were filtered by spatio-temporal proximity between the Φ -sat-2 and Sentinel-2 overpasses and by a cloud-cover threshold on both acquisitions, and precise pixel-level alignment was achieved using the LightGlue deep feature-matching algorithm [36]. This algorithm was chosen for its ability to identify corresponding semantic keypoints across the two sensors despite their differing spatial and radiometric characteristics.

The choice of Sentinel-2 as the paired source sensor follows from the two instruments' band layouts. Figure 3.2 compares their spectral response functions and their per-band spatial resolutions. Across 400–900 nm each Φ -sat-2 multispectral band has a Sentinel-

sensor’s characteristics from an existing labeled source (typically Sentinel-2) has become a standard tool for pre-mission model development [31] [32]. The simulator developed for the ESA Φ -lab OrbitalAI challenge [2] models this transformation as a sequence of physical operations in radiance space, calibrated against real flight data from related missions: a spectral and spatial resampling to the target sensor’s band configuration and ground sampling distance, a band-to-band misalignment consistent with the push-broom acquisition geometry, a signal-to-noise degradation, and an optical blur modeled through the sensor’s modulation transfer function. This simulator underlies both the degradation pipeline used in this thesis and, independently, the pre-training and benchmarking datasets used by Φ satNet (Section 2.6). It is a shared and validated infrastructure within this research context, and neither work claims it as its own contribution. Goudemant et al. [33] built an on-board marine anomaly detector for oil spills, algal blooms and sediment flooding, using the same OrbitalAI simulators to generate representative Φ -sat-2 and IMAGIN-e imagery from Sentinel-2 L1C scenes of documented pollution events. Their processing chain is the one described above: a conversion to top-of-atmosphere radiance, a Φ -sat-2-specific panchromatic band synthesised from the multispectral bands by a spectral-response-weighted combination, a spatial resampling to the target ground sampling distance, a band-to-band misalignment, a signal-to-noise degradation, a modulation transfer function blurring, and a conversion back to reflectance. They trained on simulated Φ -sat-2 patches and deployed the models on the same Intel Myriad 2 accelerator this thesis targets, which makes their work a useful independent confirmation that the simulator can support a full development cycle from a mission that has not yet delivered labelled data.

This study also illustrates the limit of the approach. Its performance was established on simulated imagery, and its authors identify two pieces of work still to be done: validating the system on in-flight data, and sensitivity testing against individual sensor phenomena such as noise and spectral deregistration. The simulator makes the pre-launch development possible, but it cannot establish how well the simulation stands in for the real sensor. This is the question the co-registered real acquisitions of Chapter 3 are assembled to answer.

2.5 Knowledge Distillation Across Architectures

Knowledge distillation (KD) [14] compresses a large “teacher” model into a smaller “student”, by training the student to reproduce the teacher’s output distribution. By transferring the teacher’s semantic knowledge, KD allows the student to generalize better on large datasets, and with a lower computational cost. Some feature-based variants exist that use the intermediate representations of the teacher, instead of training the student only against hard labels [34]. Le et al. [35] proposed a distillation method between a ViT teacher and a ResNet student for image classification, which uses a dynamic weighting mechanism to adjust each teacher’s influence on the student, based on the teacher’s confidence in its predictions. A cross-architecture distillation is a necessary step here, since ViT inference is not supported on the Φ -sat-2 on-board hardware (Section 2.1). On top of it, this thesis gives the student itself a domain-adaptation mechanism (the domain-specific batch normalization of Section 3.5). The student therefore learns the domain-specific characteristics itself, instead of inheriting whatever invariance the teacher happens to have learned. A paired-contrastive alignment objective exploiting the

2.3 Domain Gap

A domain gap is a distribution shift between the data a model was trained on and the data it has to work with once it is deployed. Because the model holds weights modeling distributions of the source domain, such a shift generally translates to a loss of accuracy, even when nothing about the task has changed. Ben-David et al. [13] formalize this intuition with a bound on the target-domain error, which they limit by the sum of three terms: the error on the source domain, a divergence between the two marginal input distributions, and the error of the best hypothesis on both domains jointly. Only the second of these describes the gap itself, and it is the term any adaptation method has to act on.

This problem is particularly relevant in on-board AI, where the model is fixed at uplink time while the acquisitions keep varying with the illumination, the atmosphere, the season, the geography and the behaviour of the instrument itself. It is useful to separate three kinds of shift, since they have different causes and can be minimized in different ways [26].

- *Covariate shift* is a change in the distribution of the inputs while the labelling rule stays the same: the same scene content is represented differently, for instance because the imagery carries blur, sensor noise or haze that the training data did not.
- *Prior shift* is a change in the distribution of the labels themselves, as when a class that is well represented in the source domain is rare or absent in the target, so a model calibrated on the source over-predicts it.
- *Concept shift* is a change in what a given label looks like: a forest in Brittany and a tropical forest in Vietnam carry the same land-cover label while containing different species with different spectral signatures, so the mapping from label to appearance is no longer the one the model learned.

This thesis is concerned with the covariate shift, and specifically with the part of it that is attributable to the instrument sensor. The three kinds are difficult to separate in practice, because two datasets acquired by different sensors normally also differ in where and when they were acquired, and therefore in class balance and in scene content: a measured drop in accuracy then confounds all three. The co-registered triplets described in Chapter 3 are constructed to remove that confound. Since each triplet holds views of the same ground at very nearly the same time, the label distribution and the underlying scene are held fixed across those views by construction, which leaves the sensor as the only factor that varies, and the shift it induces as the only one being measured.

2.3.1 Domain Adaptation

Domain adaptation (DA) addresses the general problem of a model trained on a source distribution being applied to a target distribution with different input statistics but unchanged semantics [26]. In remote sensing applications, when target data is available, there are two main types of DA: unsupervised (UDA), where the target data is unlabeled, and supervised (SDA), where a small amount of labeled target data is available. Within satellite on-board AI, Du et al. [27] demonstrated DA for onboard multispectral

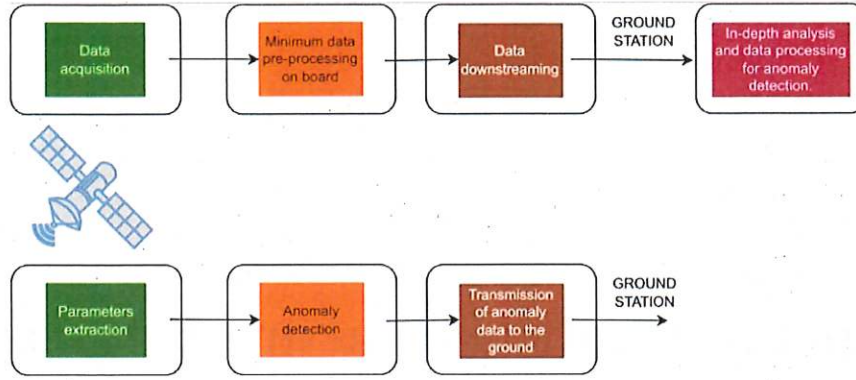


Figure 2.1: Top Workflow (Bent-Pipe Architecture): Data is acquired and minimally preprocessed on the satellite before being transmitted to the ground station, where in-depth analysis for anomalies is conducted. Bottom Workflow: The satellite extracts parameters and detects anomalies onboard, transmitting only the relevant anomaly data to the ground station, reducing data transmission and enabling faster detection. From Razzano et al. [1].

cloud-detection network on board a hyperspectral imager, and it established on-board AI as a reliable and accurate tool for that task [16]. Two constraints shaped the result. The first one was the on-board hardware: a dedicated processing had to be developed for the mission, operating within the power budget of the platform and tolerating the faults that radiation induces in orbit. The second one was the lack of data, since there wasn't enough imagery from the HyperScout product line to train a network to label every pixel as cloudy or clear. This is a recurring difficulty for space imagers, since a bespoke camera is generally developed for each new mission and its archive necessarily begins empty. The network eventually deployed was a custom segmentation architecture designed around the limits of the accelerator, and the bound on intra-layer memory in particular governed how its convolutional and deconvolutional layers could be implemented [17].

CNN-based models of this kind are efficient, but their locality leaves them less able to capture the long-range patterns and the contextual dependencies that more demanding remote sensing tasks require. ViT architectures have become popular in computer vision because they capture the global context through self-attention, and they often outperform CNNs on a variety of tasks [18]. This makes ViT-based models strong candidates for on-board EO applications, and in a comparative study of ViT variants for onboard satellite image classification, Le et al. [19] report that pre-trained ViTs consistently outperform equivalent models trained from scratch. These pre-trained models also cost less to compute and are more resilient to noise, which is close to the combination an on-board setting needs the most. Their cost nonetheless remains the primary barrier to deployment, since the large parameter counts and the attention computation that give ViTs their advantage are precisely what a resource-constrained satellite can least afford [20]. The hardware on board Φ -sat-2 is a key constraint on what can be deployed there, as the CogniSAT-XE1 payload carries the Intel Movidius Myriad 2 VPU [8], whose support ended in 2020, which makes it legacy hardware. The Φ satNet effort reports that the most recent OpenVINO release supporting this hardware is version 2020.3, that experimental testing confirmed only ONNX opset 8–11 are available¹ and that the resulting lack of

¹Supported operations: https://docs.openvino.ai/2023.3/openvino_docs_ops_opset11.html

Answering these questions would show that the generalization promise of large **GeoFMs**, which is a broad semantic transfer from limited labels, is compatible with the low-latency and resource-constrained inference that on-board emergency-response applications require, provided that the measured domain gap is addressed.

hardware.

This thesis addresses both obstacles at once, with TerraMind and Φ -sat-2 as the case study, and flood segmentation as the motivating downstream application. The approach combines three elements. First, a physics-based sensor degradation pipeline, based on the simulator developed for the ESA Φ -lab OrbitalAI challenge[2], processes Sentinel-2 imagery to simulate the Φ -sat-2 sensor characteristics. This gives us labeled training data for a target sensor that hasn't accumulated data for flood segmentation yet. Second, a triplets dataset of spatially and temporally co-registered acquisitions from real Φ -sat-2, real Sentinel-2 and simulated Φ -sat-2 enables us to measure the domain gap and to test adaptation mechanisms against it. Third, domain-specific batch normalization[4] is applied during the knowledge distillation[14] of the TerraMind encoder into a lightweight CNN. With this method, the deployable model adapts to the sensor its getting data from at inference. A paired-contrastive alignment objective exploiting the co-registered triplets directly was also implemented and evaluated as an alternative mechanism, but it did not improve downstream task performance beyond what domain-specific normalization already provides.

1.1 Problem Definition

The main technical problem is a covariate shift between the sensor domain a GeoFM is pre-trained on and the sensor domain it is tasked to operate on. Let X denote the space of satellite image patches and Y the space of semantic labels (land-cover class, flood/background). Writing p_s and p_t for the source (pre-training, e.g. Sentinel-2) and target (operational, e.g. Φ -sat-2) distributions, a covariate shift is present when

$$p_t(x) \neq p_s(x) \quad \text{while} \quad p_t(y | x) = p_s(y | x). \quad (1.1)$$

In practice, the physical scene and its semantic content are identical across the two domains, and it is the sensors that produce different pixel statistics. The causes are the differing spectral response functions, the point spread functions, the signal-to-noise characteristics and the geometric band-to-band alignment, which are all properties of the sensor rather than properties of the scene itself. A model whose learned representation is sensitive to these sensor-specific statistics instead of the underlying semantics will therefore degrade on the target domain, even though nothing about the task itself has changed.

Prior work on the GeoFM domain gap has usually measured this shift only indirectly, through downstream task performance, and on source and target datasets that differ in geography, acquisition date and scene content in addition to the sensor. The sensor-specific shift this thesis is concerned with is then confounded with the shift due to the underlying data distribution itself. The co-registered triplet dataset used in this work is designed specifically to remove that confound, since the real Φ -sat-2, real Sentinel-2 and simulated patches used for evaluation depict the same scene at the same time.

1.2 Research Gap

Two related works approach the gap this thesis addresses in similar ways. Du et al. [15] demonstrated an on-orbit GeoFM (compressed Prithvi) on the Kanyini CubeSat and the IMAGIN-e payload, and adapted it to the target sensor by retraining task heads on

PAD	Proxy $\mathcal{A}$ -Distance
PSF	Point Spread Function
RE	Red Edge (spectral band)
RGB	Red, Green, Blue
RTC	Radiometric Terrain Corrected
SDA	Supervised Domain Adaptation
SNR	Signal-to-Noise Ratio
TOA	Top of Atmosphere
UDA	Unsupervised Domain Adaptation
UMAP	Uniform Manifold Approximation and Projection
ViT	Vision Transformer
VPU	Vision Processing Unit

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Abstract

On-board processing is a promising answer to the downlink latency and bandwidth constraints that limit the usefulness of Earth observation data for time-critical applications such as flood and wildfire response. Geospatial Foundation Models (GeoFMs) are interesting candidates for this task, since their large-scale multimodal pre-training gives semantic feature extractors that generalize across tasks with few labeled examples. However, these models are pre-trained on archived data from missions such as Sentinel-2, and their behaviour once they are exposed to a new sensor is not known. This thesis studies three questions, with TerraMind, a Vision Transformer (ViT)-based GeoFM and the European Space Agency’s Φ -sat-2 CubeSat mission as the case study: whether TerraMind produces domain-invariant latent representations on real Φ -sat-2 imagery, whether a lightweight Convolutional Neural Network (CNN) distilled from it can be made sensor-adaptive, and whether doing so is better than simply training the compact model on the target sensor.

We use a co-registered triplet dataset of real Φ -sat-2, real Sentinel-2 and physics-simulated Φ -sat-2 imagery, which lets us measure the domain gap directly, as the distance in TerraMind’s embedding space between two sensor views of the same ground. The gap is already present in the early layers of the encoder, and a linear probe transfers only a third of its source-domain accuracy to the target sensor. This points to a low-level, radiometric origin in the two sensors’ differing spectral response functions, since a semantic cause would rather have surfaced in the deeper layers. Plain fine-tuning on physics-degraded data does not close the gap either. Domain-specific batch normalization, applied to the student during knowledge distillation into a lightweight CNN does reduce it (cosine similarity $0.098 \rightarrow 0.948$, probe transfer $33\% \rightarrow 97\%$), and it only requires the sensor of each input to be known. A paired-contrastive objective exploiting the co-registered pairing was also implemented and evaluated, but it did not do better than domain-specific normalization alone. Whether this adaptation is worth adopting then depends on the label budget. The distilled, domain-adapted student outperforms a supervised baseline of identical architecture trained directly on the target sensor at a representative labeled-data budget (mIoU 0.4532 vs. 0.3989), but it loses to that same baseline at a smaller one (mIoU 0.2679 vs. 0.2916). A few-shot comparison against Φ satNet, an on-orbit GeoFM developed for the same mission, is planned but remains to be run.